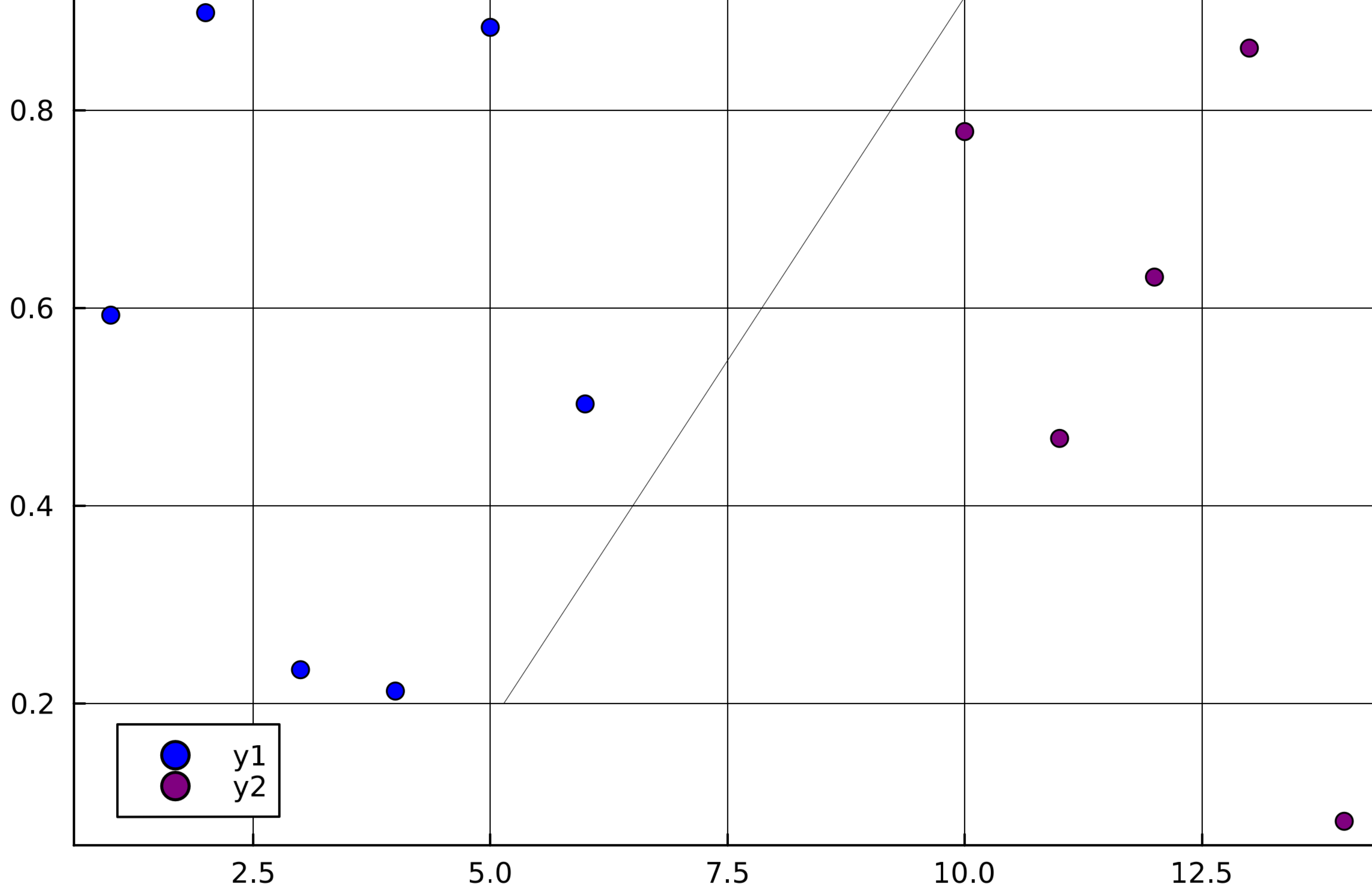


ta911s

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PhD Computer Science



unsupervised learning - part 1



- what if the labels were omitted?

goal

- discover interesting patterns in the observations
 - analyse and group unlabelled data
 - discover subgroups among the observations

clustering

- group unlabelled data based on similarities or differences
- types of algorithms
 - centroid-based
 - density-based
 - distribution-based
 - hierarchical

- more classification
 - membership
 - exclusive vs multiple clusters
- pre-defined numbers
 - hard vs soft
- dataset capacity
 - small vs medium vs large

procedure

- prepare data
- choose distance/similarity metric
- run clustering algorithm
- evaluate, analyse and improve

distance & similarity

Minkowski distance

$$\left(\sum_{\ell=1}^d |x_{i\ell} - x_{j\ell}|^n \right)^{\frac{1}{n}}$$

Euclidian distance

$$\left(\sum_{\ell=1}^d |x_{i\ell} - x_{j\ell}|^2 \right)^{\frac{1}{2}}$$

standardized Euclidian distance

$$\left(\sum_{\ell=1}^d \left| \frac{x_{i\ell} - x_{j\ell}}{s_{\ell}} \right|^2 \right)^{\frac{1}{2}}$$

cosine distance

$$1 - \cos(\alpha) = \frac{\mathbf{x}_i^T \mathbf{x}_j}{|\mathbf{x}_i| |\mathbf{x}_j|}$$

- more
 - Pearson correlation distance
 - Mahalanobis distance

Jacquard similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

algorithms

k-means

$C_1, \dots, C_k$ denote sets containing the indices of obs.

$$C_1 \cup C_k \cup \dots \cup C_k = \{1, \dots, n\}$$

$$C_1 \cap C_k \cap \dots \cap C_k = \emptyset$$

$$\underset{C_1, \dots, C_K}{\text{minimize}} \quad \sum_{k=1}^K \mathbf{WCV}(C_k)$$

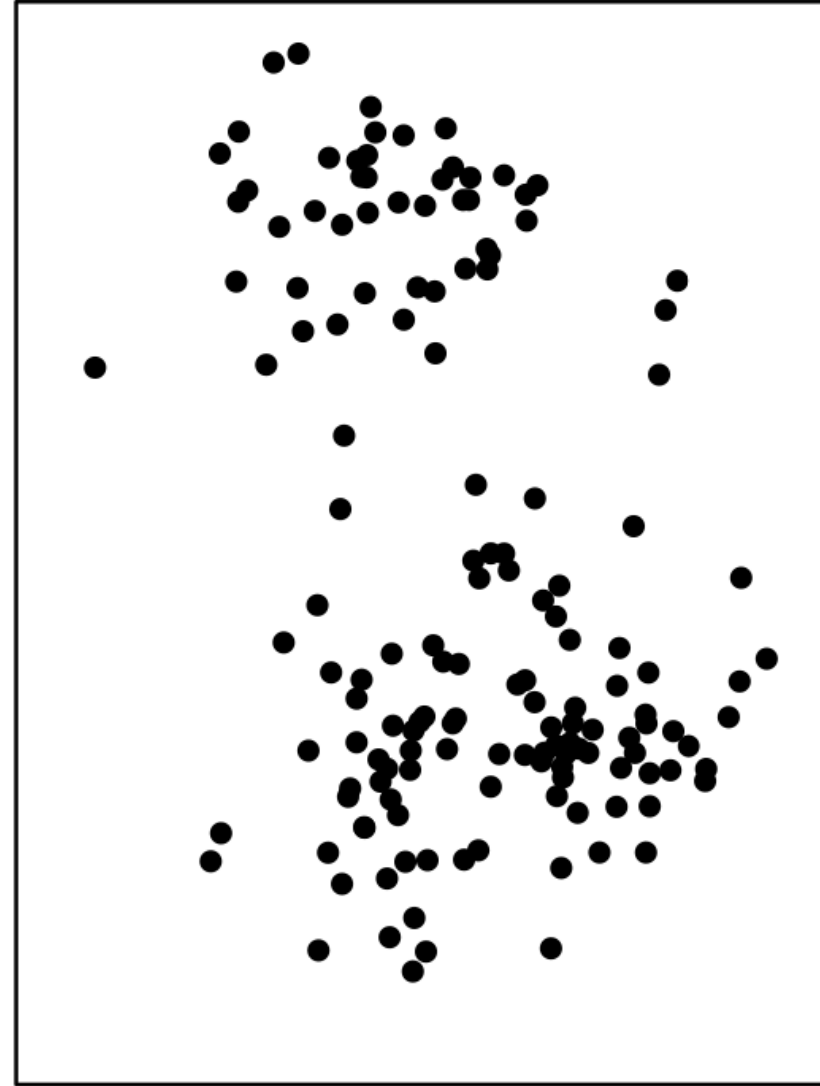
$$\mathbf{WCV}(C_k) = \frac{1}{|C_k|} \sum_{l, l' \in C_k} \sum_{j=1}^p (x_{lj} - x_{l'j})^2$$

euclidian distance

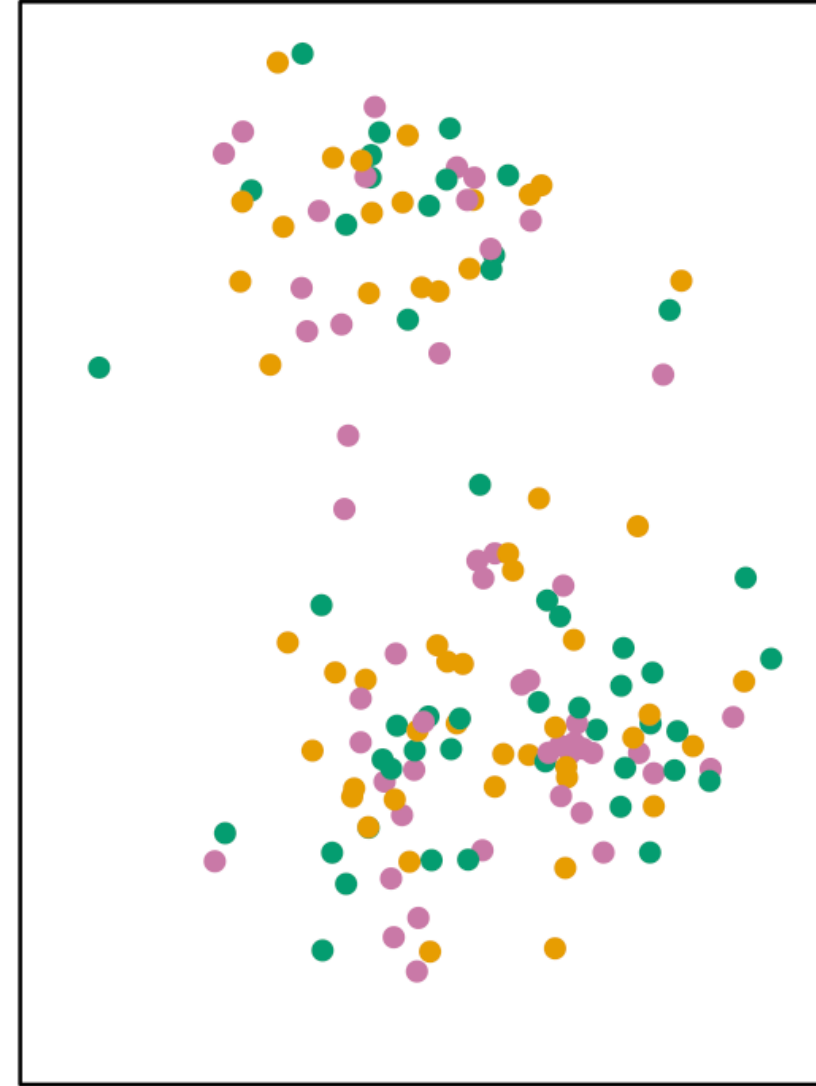
steps

- randomly assign a number from 1 to **K** to each observation
- iterate until the cluster assignment does not change
 - compute the cluster centroid for each cluster
 - assign each observation to the closest centroid

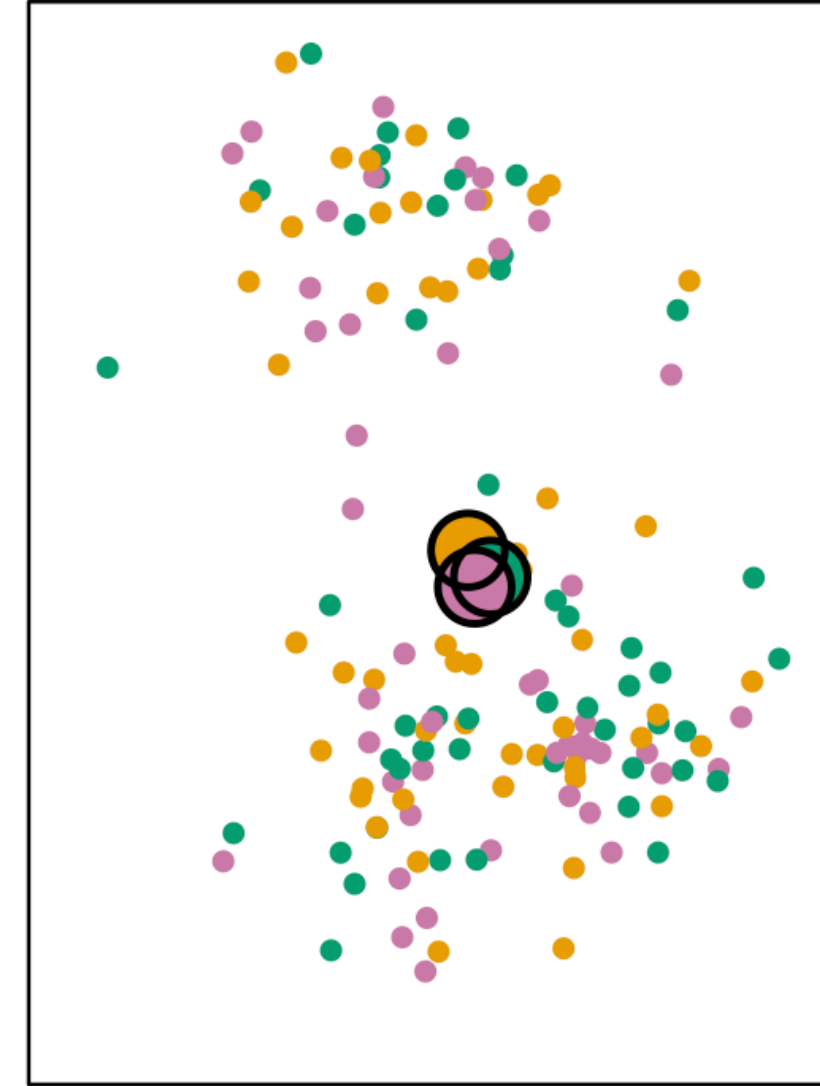
Data



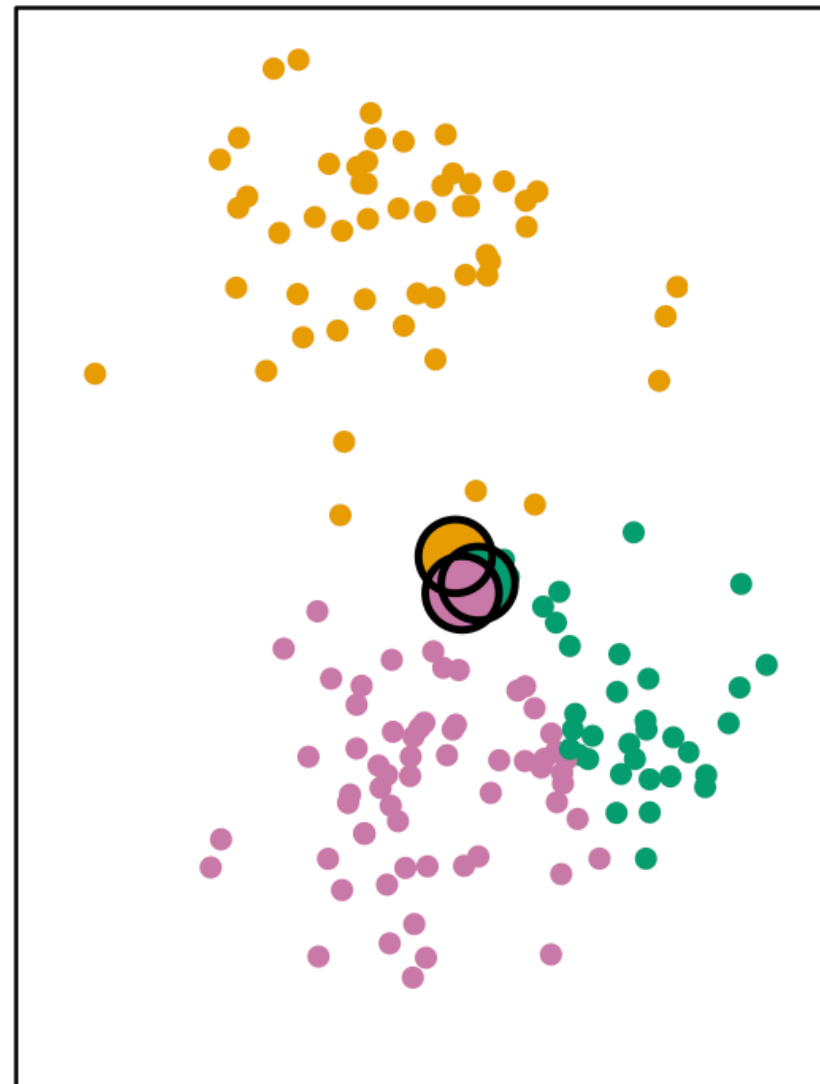
Step 1



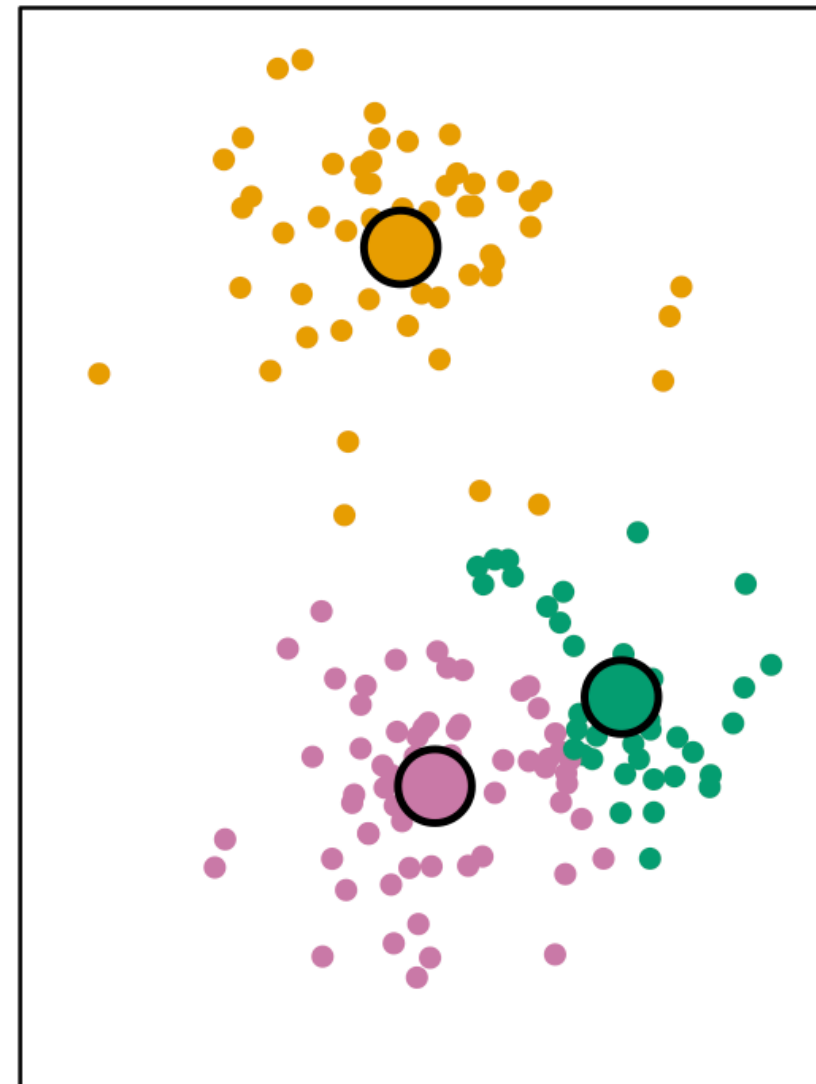
Iteration 1, Step 2a



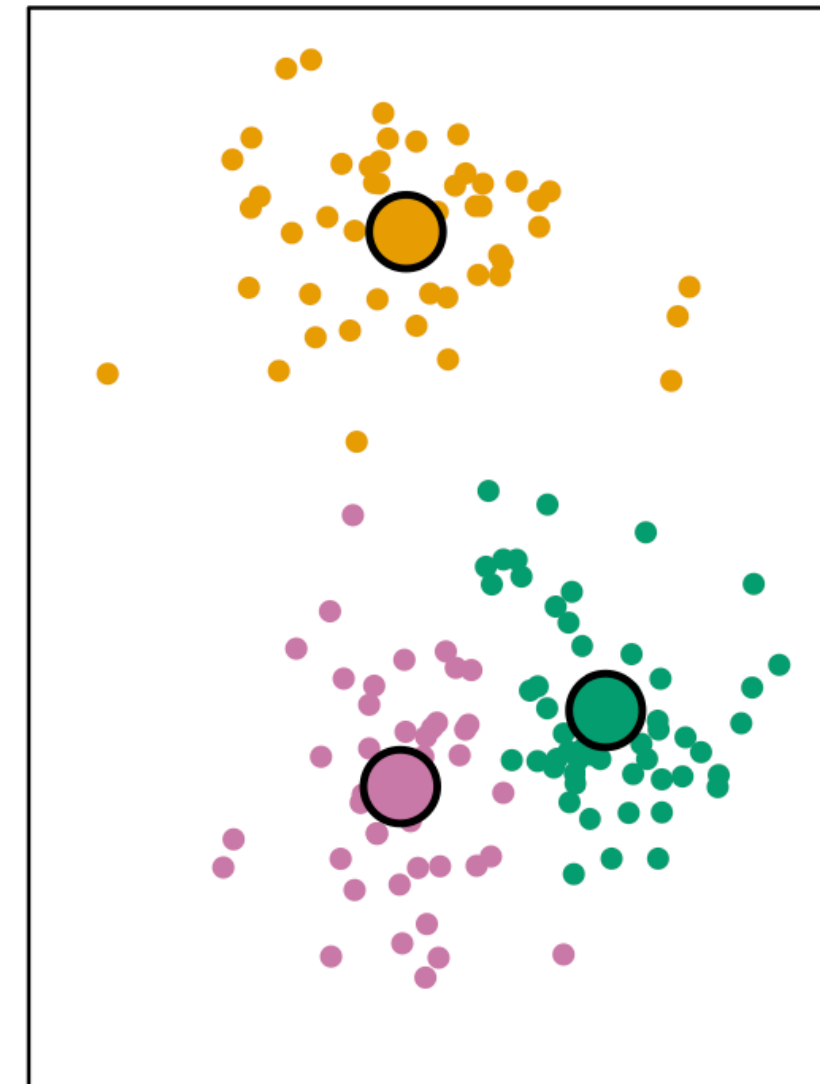
Iteration 1, Step 2b



Iteration 2, Step 2a



Final Results

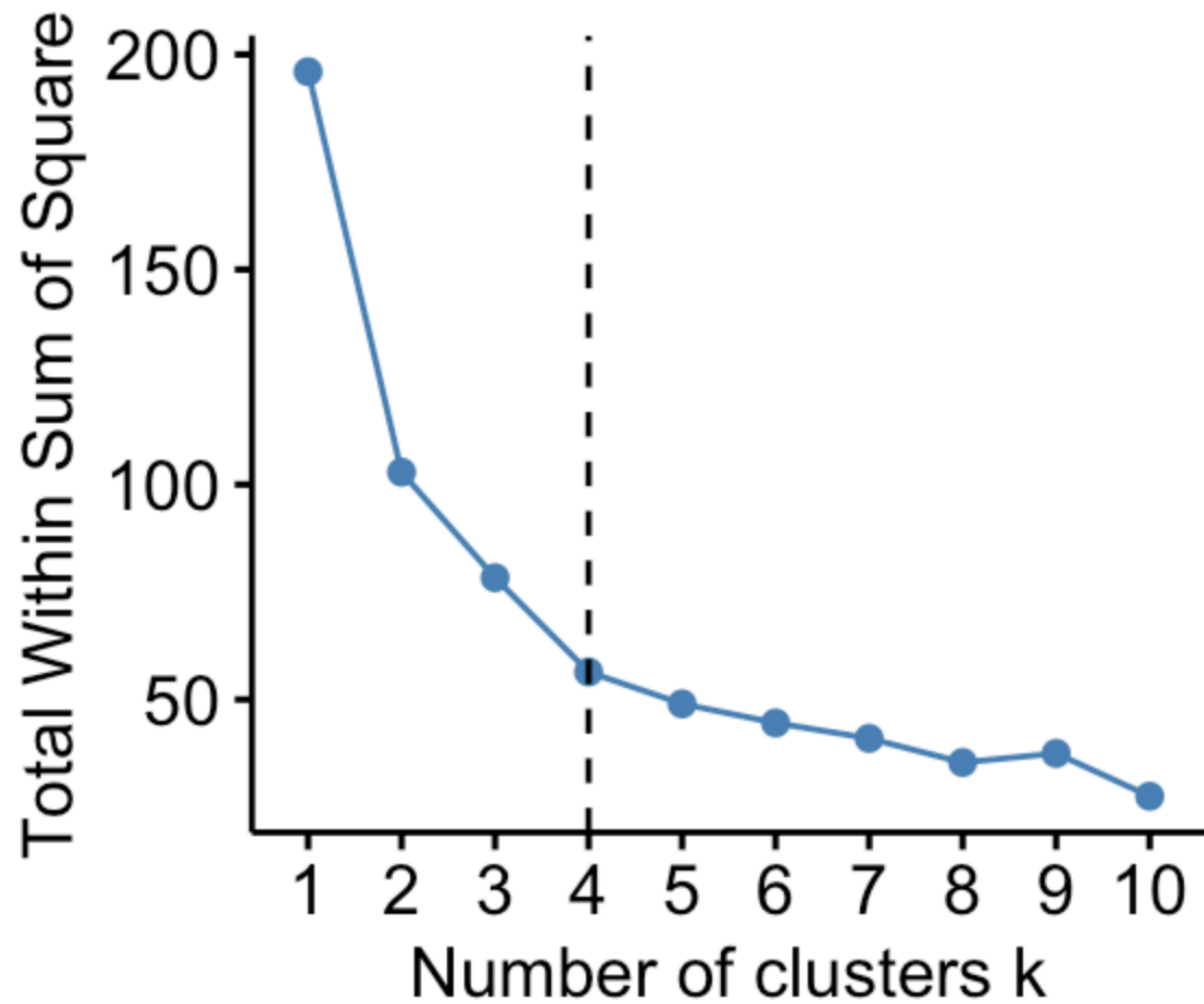


- initial centroids
 - affect quality of clustering
 - run several rounds

- how is K selected?
 - elbow method
 - vary the value of k and plot error
 - optimal k is the “**elbow**” of the curve

Optimal number of clusters

Elbow method



- silhouette score

$$s(\iota) = \frac{\frac{1}{|C(\iota)| - 1} \sum_{C(\iota), \iota \neq J} d(\iota, J) - \min_{\iota \neq J} \left(\frac{1}{|C(J)|} \right) \sum_{J \in C(J)} d(\iota, J)}{\max \left(\frac{1}{|C(\iota)| - 1} \sum_{C(\iota), \iota \neq J} d(\iota, J), \min_{\iota \neq J} \left(\frac{1}{|C(J)|} \right) \sum_{J \in C(J)} d(\iota, J) \right)}$$

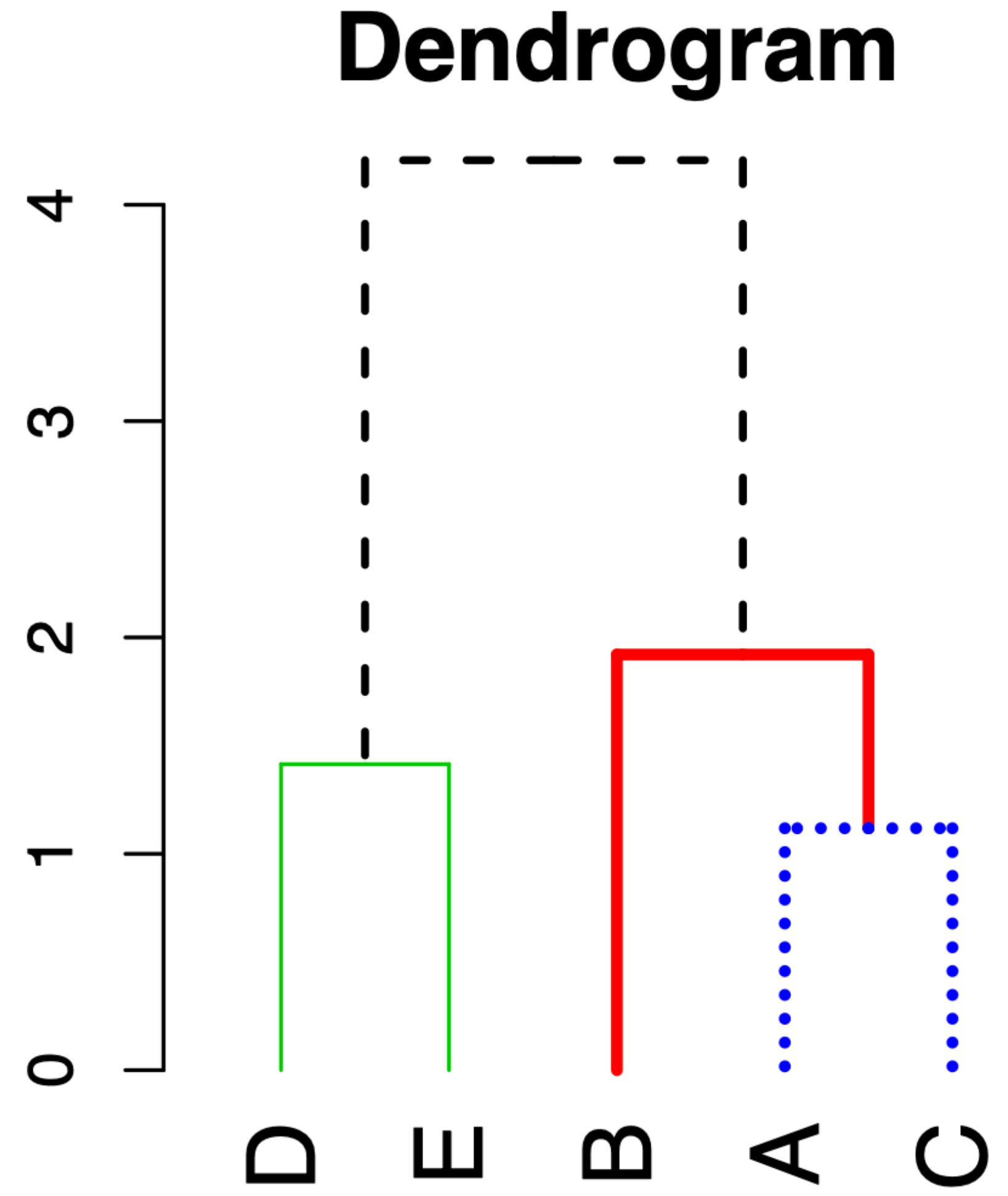
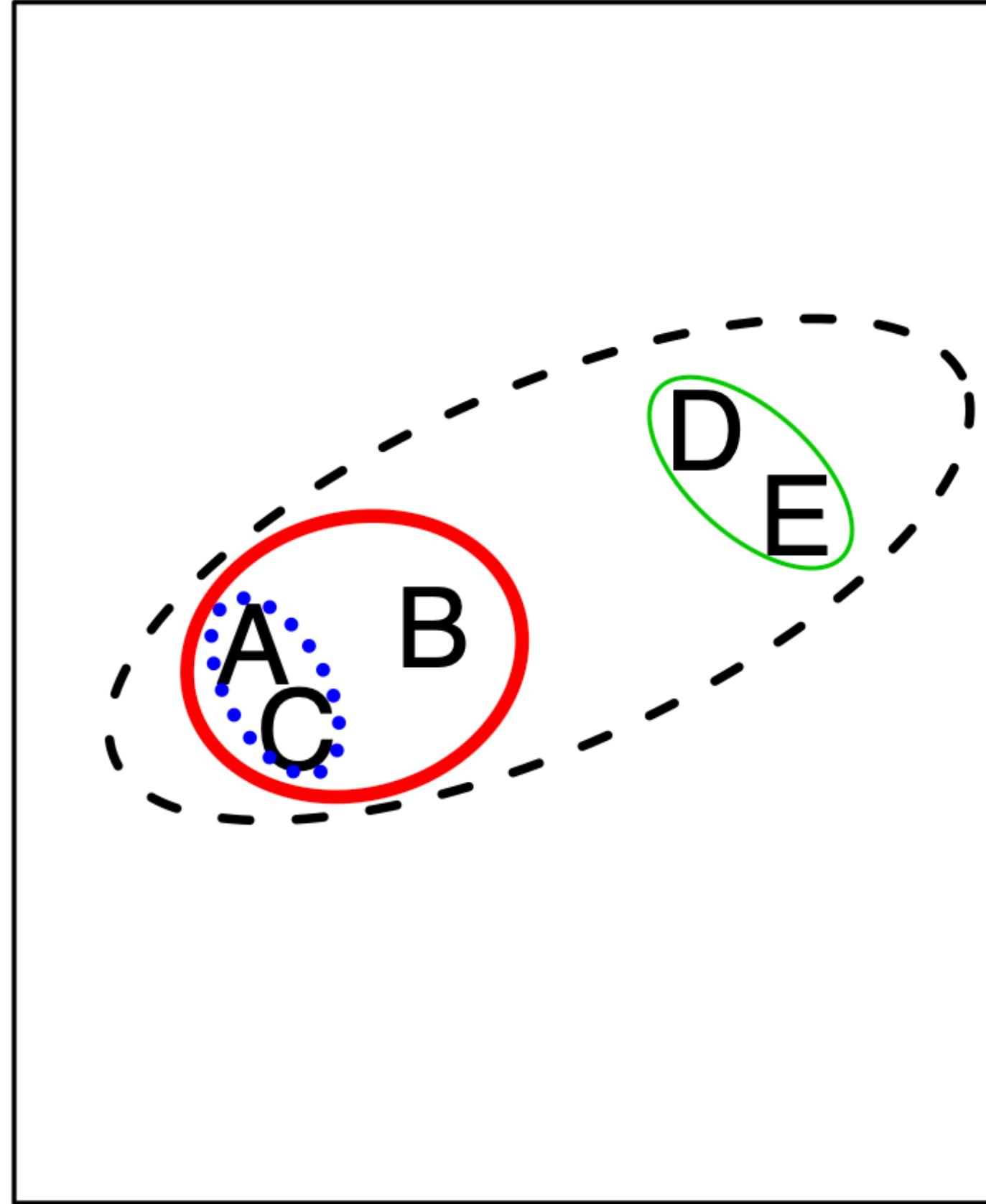
agglomerative clustering

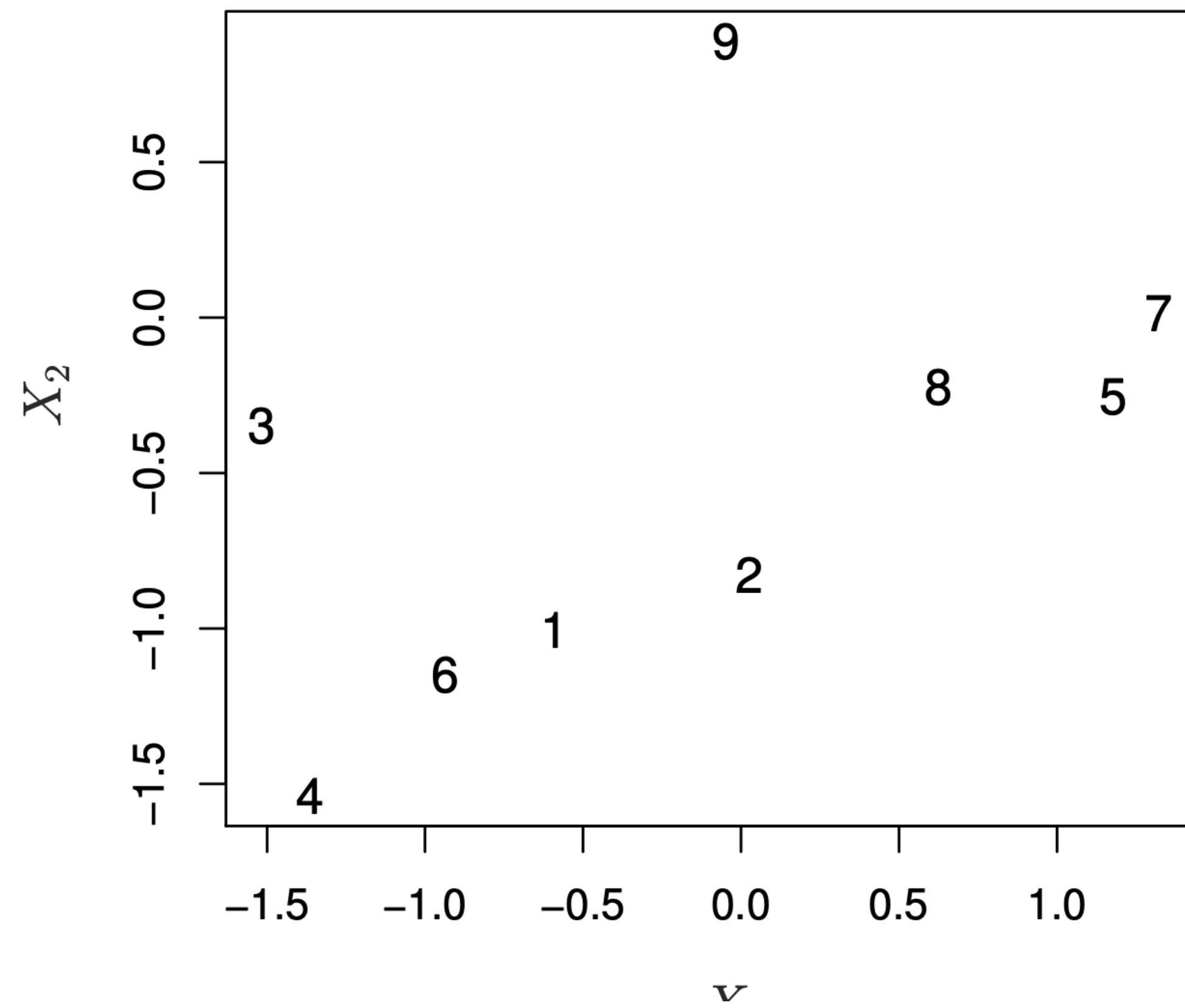
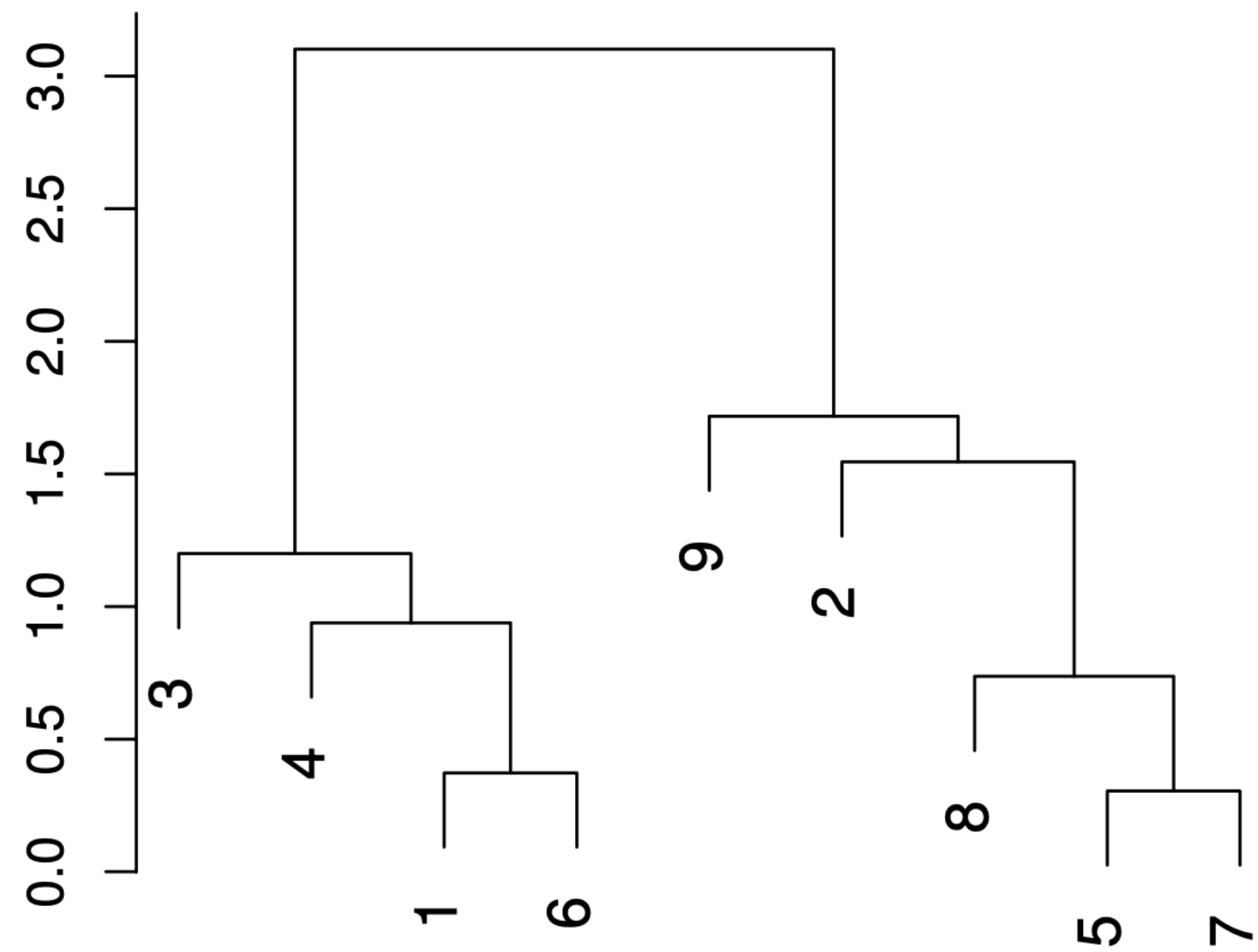
- build a hierarchy in a bottom-up fashion

Steps

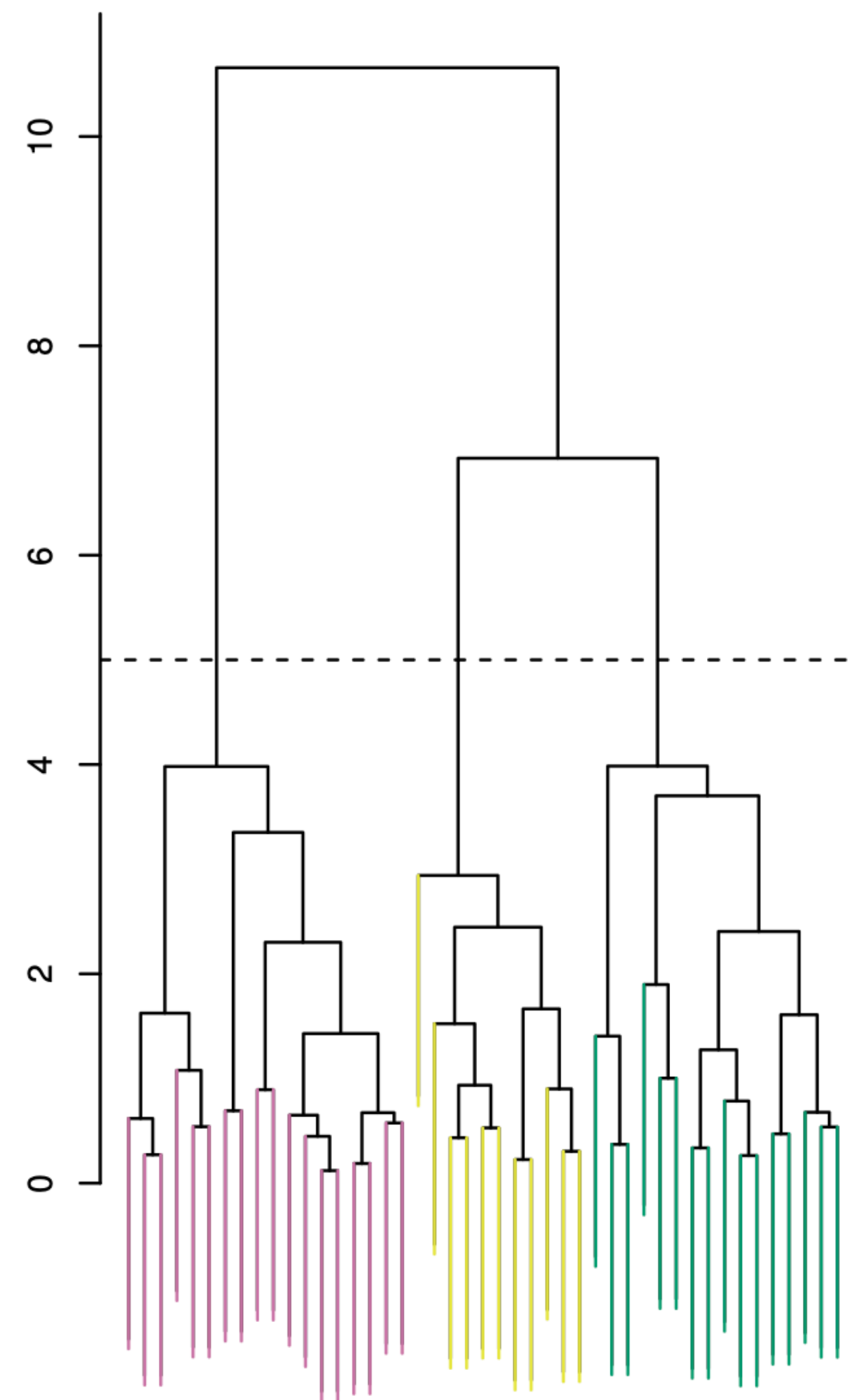
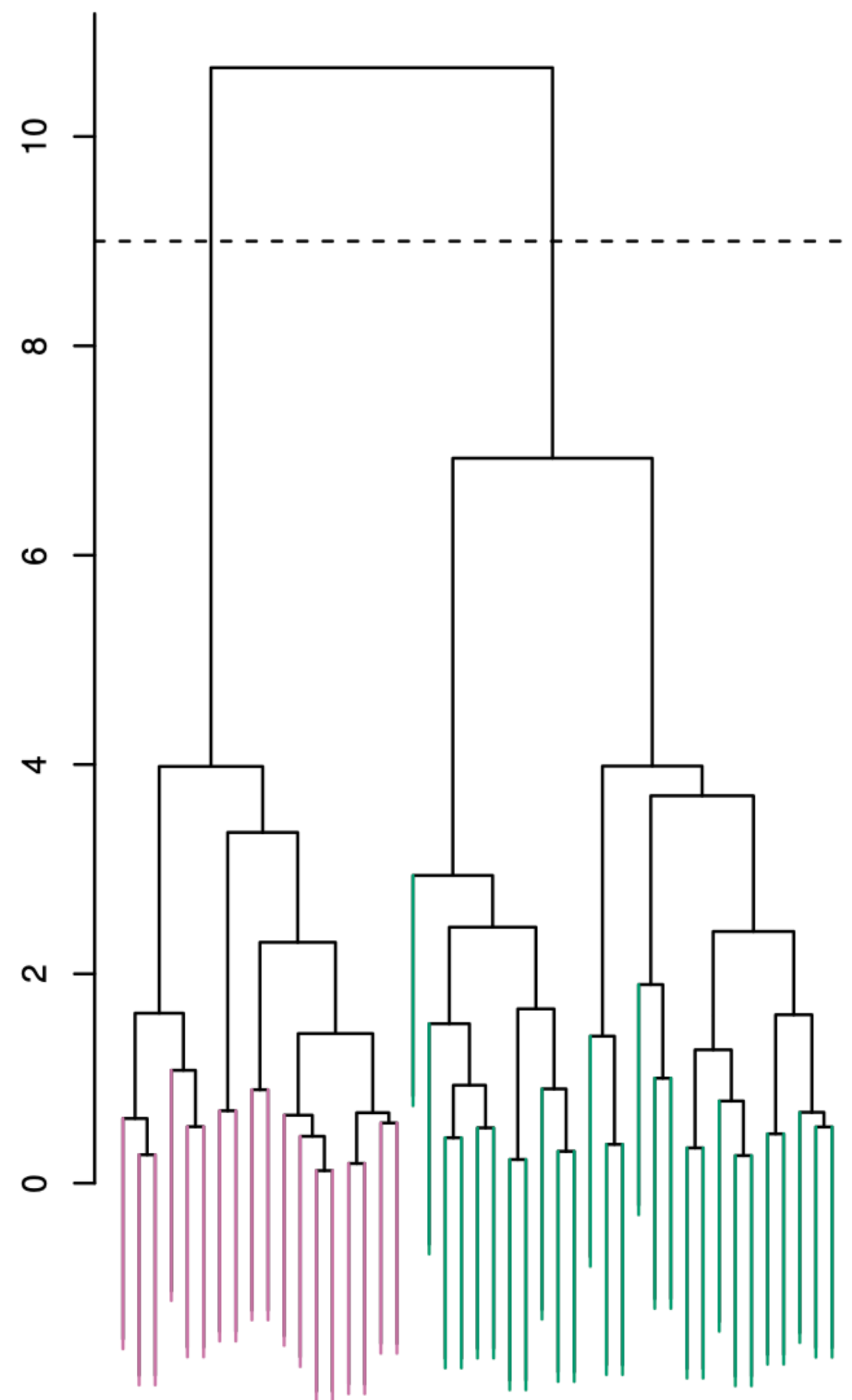
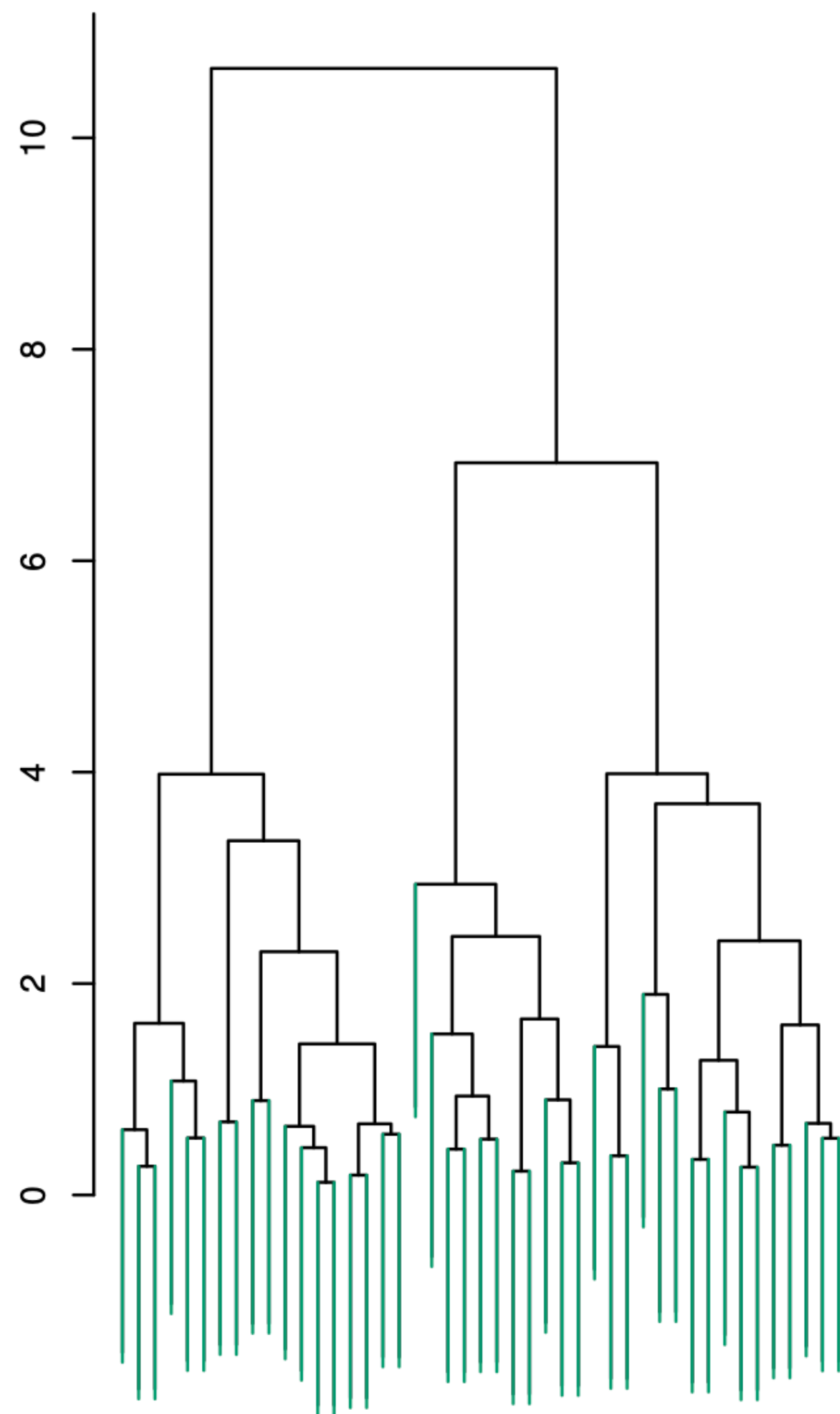
- start with each data point in its own cluster
- identify the closest two clusters and merge them
- repeat the process
- end when all data points are in a single cluster

- result is a dendrogram





- are 9 and 2 close?



- dissimilarity between clusters

- use a linkage
 - complete (largest pairwise dissimilarity)
 - single (smallest pairwise dissimilarity)
 - average (average pairwise dissimilarity)
 - centroid (centroid dissimilarity)